

DATA ANALYTICS – PROJECT 2

Exploratory Data Analysis (EDA)

Intern Details

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Role: Data Analytics Intern

Organization: DecodeLabs

Project: Project 2 – Exploratory Data Analysis (EDA)

Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the cleaned dataset to identify patterns, trends, and potential outliers. EDA helps in understanding the characteristics of the dataset and provides valuable insights for decision-making.

Dataset Overview

The cleaned dataset from Project 1 was used for analysis. The dataset contains transactional information including products, payment methods, coupon usage, order values, and order dates.

Total Records Analyzed: **1200**

Summary Statistics

Metric	Value
Total Orders	1200
Average Order Value	1053.97
Maximum Order Value	3456.40

Observation

The average order value was ₹1053.97, while the highest order value recorded was ₹3456.40.

Screenshot:

C	D	E	F	G
	Summary Statistics :			
	Count of Orders	1200		
	Average Order Value	1053.97		
	Maximum Order Value	3456.4		

Product Analysis

A Pivot Table was used to identify the most frequently ordered products.

Product Order Count

Printer 181

Tablet 179

Chair 178

Laptop 173

Desk 170

Observation

Printer was the most frequently ordered product with 181 orders, followed closely by Tablet and Chair.

Screenshot:

	A	B	C
1	PIVOT TABLES :		
2			
3	Row Labels	Count of Product	
4	Printer	181	
5	Tablet	179	
6	Chair	178	
7	Laptop	173	
8	Desk	170	
9	Monitor	163	
10	Phone	156	
11	Grand Total	1200	

Payment Method Analysis

A Pivot Table was created to analyze customer payment preferences.

Payment Method Order Count

Online 258

Cash 246

Credit Card 234

Debit Card 232

Gift Card 230

Observation

Online payment was the most preferred payment method among customers.

Screenshot:

13	Row Labels	Count of PaymentMethod
14	Online	258
15	Cash	246
16	Credit Card	234
17	Debit Card	232
18	Gift Card	230
19	Grand Total	1200

Coupon Usage Analysis

A Pivot Table was used to evaluate coupon code usage.

Coupon Code Usage Count

FREESHIP 313

No Coupon 309

WINTER15 292

SAVE10 286

Observation

FREESHIP was the most frequently used coupon code. A significant number of customers also placed orders without using any coupon.

Screenshot:

21	Row Labels	Count of CouponCode	
22	FREESHIP	313	
23	No Coupon	309	
24	WINTER15	292	
25	SAVE10	286	
26	Grand Total	1200	

Outlier Detection

The TotalPrice column was analyzed to identify unusually high order values.

Metric	Value
Average Order Value	1053.97
Maximum Order Value	3456.40

Observation

The maximum order value was significantly higher than the average order value, indicating the presence of potential high-value outliers in the dataset.

Tools Used

- Microsoft Excel
- Pivot Tables
- Descriptive Statistics
- Sorting and Filtering

Key Insights

1. A total of 1200 orders were analyzed.
2. The average order value was ₹1053.97.
3. Printer was the most ordered product.
4. Online payment was the most preferred payment method.
5. FREESHIP was the most commonly used coupon code.
6. A few high-value orders were identified as potential outliers.

Conclusion

Exploratory Data Analysis was successfully performed on the cleaned dataset. Key business insights were identified using summary statistics and Pivot Tables. Product demand, payment preferences, coupon usage patterns, and potential outliers were analyzed effectively. The findings provide a clear understanding of customer behavior and transaction trends within the dataset.